FS Task

IMPORTANT:

THIS TASK MUST BE COMPLETED EXCLUSIVELY BY YOU. NO EXTERNAL HELP OF ANY KIND IS ALLOWED.

IN CASE THESE REQUIREMENTS ARE NOT MET, WE RESERVE THE RIGHT TO WITHHOLD PAYMENT.

Stack Requirements:

1. Angular Version 19
2. AG Grid C AG Charts (33.0)
3. Angular Material
4. Angular Material Icons
5. Node v22

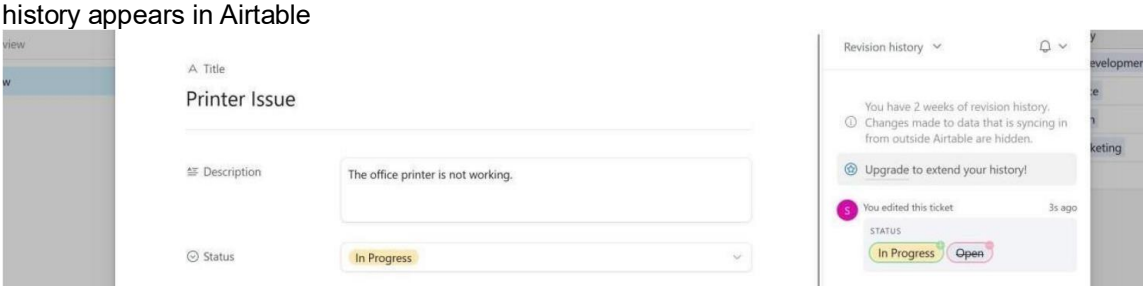
Technical Details:

Part A:

- We need to create an integration for Airtable
- Here is the documentation: https://airtable.com/developers/web/api/introduction
- Build OAuth Authentication.
- We need to fetch the following endpoints: (BaseURL: https://api.airtable.com/v0)
- Projects (bases) : /meta/bases
- Tables (tables): /meta/bases/${baseId}/tables
- Tickets (pages): /${baseId}/${tableId}
- Users: /Users
- We only need to focus on the above endpoints and store all the pages in a separate MongoDB collection. (Important: Make sure to use Airtable API Pagination)

Part B:

- Once we can fetch all the above endpoints, we need to write a Custom Scraping Method for changelogs.
- We are looking to fetch **Revision History**. Refer to the screenshot below to see how the revision history appears in Airtable



- We need to write a custom scraping method that should work as follow:
  - We need to automatically retrieve the Cookies from Airtable so we can pass the cookies to the Revision History endpoint (i.e /readRowActivitiesAndComments)
  - We need to ability to check if the cookies are valid.
  - We need to fetch Revision History for each task that we have fetched earlier using the API.
  - The Revision History response is in HTML, we need to parse the response into the following format before storing it to the database:

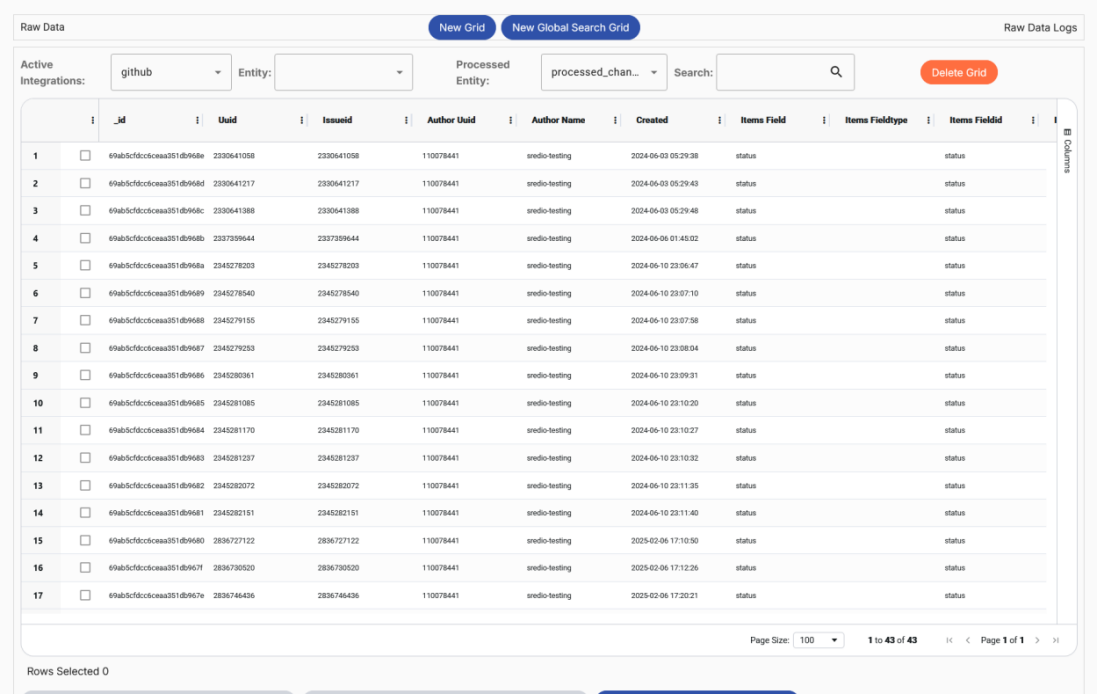
```
{
    uuid: activityId,
    issueId: ticketId,
    columnType: columnType,
    oldValue: oldValue,
    newValue: newValue,
    createdDate: new Date(activityData.createdTime),
    authoredBy: activityData.originatingUserId,
}
```

- Once we are able to fetch the Revision History, store it in the database.
- You will need to test this with atleast 200 Pages and make sure the revision history is retrieved.
- **Important: In the Revision History: We are looking for Assignee and Status Changes**

Status	Assignee
In Progress	SRED.io Integration
In Progress	SRED.io Integration
Closed	
Closed	

- We need the Custom Scraping Method to automatically retrieve the cookies and pass it to the Airtable Integration Service to get the Revision History for each ticket (page).
- It is important to check if the cookies are valid or not. This can be done by making a request to the endpoints and if the cookies are expired we need to redo it.
- We also need to have the ability to pass the MFA Code from the frontend so we can easily get the cookies and run the Custom Scraping Method.

Part C:



	_id	Uuid	Issueid	Author Uuid	Author Name	Created	Items Field	Items Fieldtype	Items Fieldid
1	69ab5cfdcc6ceaa351db968e	2330641058	2330641058	110078441	sredio-testing	2024-06-03 05:29:38	status		status
2	69ab5cfdcc6ceaa351db968d	2330641217	2330641217	110078441	sredio-testing	2024-06-03 05:29:43	status		status
3	69ab5cfdcc6ceaa351db968c	2330641388	2330641388	110078441	sredio-testing	2024-06-03 05:29:48	status		status
4	69ab5cfdcc6ceaa351db968b	2337359644	2337359644	110078441	sredio-testing	2024-06-06 01:45:02	status		status
5	69ab5cfdcc6ceaa351db968a	2345278203	2345278203	110078441	sredio-testing	2024-06-10 23:06:47	status		status
6	69ab5cfdcc6ceaa351db9689	2345278540	2345278540	110078441	sredio-testing	2024-06-10 23:07:10	status		status
7	69ab5cfdcc6ceaa351db9688	2345279155	2345279155	110078441	sredio-testing	2024-06-10 23:07:58	status		status
8	69ab5cfdcc6ceaa351db9687	2345279253	2345279253	110078441	sredio-testing	2024-06-10 23:08:04	status		status
9	69ab5cfdcc6ceaa351db9686	2345280361	2345280361	110078441	sredio-testing	2024-06-10 23:09:31	status		status
10	69ab5cfdcc6ceaa351db9685	2345281085	2345281085	110078441	sredio-testing	2024-06-10 23:10:20	status		status
11	69ab5cfdcc6ceaa351db9684	2345281170	2345281170	110078441	sredio-testing	2024-06-10 23:10:27	status		status
12	69ab5cfdcc6ceaa351db9683	2345281237	2345281237	110078441	sredio-testing	2024-06-10 23:10:32	status		status
13	69ab5cfdcc6ceaa351db9682	2345282072	2345282072	110078441	sredio-testing	2024-06-10 23:11:35	status		status
14	69ab5cfdcc6ceaa351db9681	2345282151	2345282151	110078441	sredio-testing	2024-06-10 23:11:40	status		status
15	69ab5cfdcc6ceaa351db9680	2836727122	2836727122	110078441	sredio-testing	2025-02-06 17:10:50	status		status
16	69ab5cfdcc6ceaa351db967f	2836730520	2836730520	110078441	sredio-testing	2025-02-06 17:12:26	status		status
17	69ab5cfdcc6ceaa351db967e	2836746436	2836746436	110078441	sredio-testing	2025-02-06 17:20:21	status		status

- Active Integrations Dropdown: Airtable
- Entity Dropdown: List of Collections in the Airtable database.
- Search: Ability to search keyword in the Easy Grid.
- Apply filter/sorting across all the columns.
- AG Grid Table will display all the fields from the collection. It will get the fields dynamically from the selected collection and display them in the AG Grid.

Note: As part of Task, it is required to build a simple UI to demonstrate the functionality. This will be a key part of the evaluation. Please ensure the UI is clean and user-friendly.

This task is designed to be interpreted as written. Make your own decisions if anything is ambiguous. Unless absolutely necessary, do not reach out to the team with questions that you can find on your own.